Reinforcement Learning for Optimal Setpoint Determination in Energy Utility Systems

Master Thesis

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Task description

Energy utility systems are the backbone of modern energy infrastructure. Their complexity ranges from small, localized systems to vast, highly interconnected infrastructure networks. Most of them operate under dynamic conditions where multiple interacting variables influence operational efficiency, cost, reliability, and stability. Conventionally, optimal setpoints for such systems are determined using mathematical optimization. These approaches rely on accurate system models and are typically formulated as constrained optimization problems. While effective for well-defined and moderately sized systems, classical optimization becomes computationally expensive as system complexity increases. Furthermore, time-varying dynamics, uncertainty from the integration of renewable sources, and model-plant mismatches makes it challenging to use purely model-based approaches.

This thesis investigates whether reinforcement learning can find optimal operating setpoints for two case studies of different complexity: a battery energy storage system and a non-linear cookie production factory. For each system, the control task is formulated as a Markov decision process (MDP) with explicit state, action, and reward definitions aligned with operational objectives. The RL agent is trained to produce adaptive setpoint policies that minimize operating cost while satisfying process constraints and preserving stable operation. The learned policies are then benchmarked against established mathematical optimization methods to determine where RL provides practical benefits and where its limitations remain.

The evaluation of the proposed method will be conducted through systematic simulation studies on two energy system case studies of varying complexity. Performance will be assessed using quantitative metrics including total operational cost, constraint violation frequency, and computational time. The results will provide a structured comparison between reinforcement learning and classical optimization and will determine under which conditions RL offers practical advantages for real-time setpoint optimization in complex energy utility systems.

Abstract

Example summary in English.

Zusammenfassung

Beispielzusammenfassung in Deutsch.

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1 Introduction

Introduces the research problem, motivation, and objectives of the thesis.

2 Theory and Background

Provides the fundamental concepts and background required to understand the systems, optimization methods, and reinforcement learning used in this work.

This section provides the theoretical basis for the control problem studied in this thesis. The central question is how operating setpoints in energy utility systems can be selected over time so that costs are reduced while physical and operational limits are respected. The same problem can be viewed from two perspectives: as a model-based constrained optimization problem, and as a sequential decision-making problem that can be addressed with reinforcement learning. The section then introduces Soft Actor-Critic (SAC), which is used in this work to learn continuous setpoint policies.

Here, RL is not treated as a replacement for physical constraints. Instead, the original constrained optimization problem is transformed into a policy-learning problem in which feasibility is encouraged or enforced through action bounds, clipping, and penalty terms. This distinction is important for the comparison with mathematical optimization, because a low-cost RL policy is only useful if its decisions remain feasible.

2.1 Setpoint Optimization in Energy Utility Systems

Defines setpoints and explains why their optimal selection is relevant for energy utility systems.

Energy utility systems are operated by choosing setpoints for controllable components such as storage units, heat pumps, valves, and bypasses. A setpoint describes the desired value of a control variable. In the systems considered here, this may be the charging or discharging power of a battery, the compressor setting of a heat pump, a mass flow rate, or a temperature level in a thermal process. These choices are not just technical details; they directly affect operating cost, energy efficiency, constraint satisfaction, and system stability.

Setpoint determination is also inherently sequential. The best operating point at one time step cannot be chosen independently of what happened before or what may be needed later. Storage systems make this especially clear. Charging a battery or a thermal energy storage can increase future flexibility, but it may also increase the current electricity cost. Discharging can reduce current grid consumption, but it leaves less stored energy for later. Good setpoint decisions therefore have to account for inter-temporal dependencies, changing boundary conditions, and physical operating limits at the same time [1, 2].

In practical operation, the controller also has to react to external signals such as electricity prices, renewable generation, production demand, and ambient conditions. Since these signals change over time, fixed rule-based strategies can become inefficient, and the operation can be formulated as a finite-horizon decision problem with time-varying boundary conditions [3]. This motivates two formulations of the same control task: constrained finite-horizon optimization and reinforcement learning.

2.2 Mathematical Optimization

Presents traditional optimization methods used for determining optimal system setpoints.

Mathematical optimization provides a natural way to describe setpoint selection. In general, optimization means selecting the best decision from a set of feasible alternatives according to a defined objective. For energy systems, the objective is often to minimize operating cost or energy consumption, while the feasible set is determined by physical equations and operational limits. A constrained optimization problem can be written as [4]:

$$\min_{x \in \Omega} f(x) \quad \text{s.t.} \quad h(x) = 0,\ g(x) \leq 0, \tag{1}$$

Here, $x \in \Omega \subseteq \mathbb{R}^n$ denotes the decision variables, $f : \mathbb{R}^n \to \mathbb{R}$ is the objective function, $h : \mathbb{R}^n \to \mathbb{R}^p$ represents equality constraints, and $g : \mathbb{R}^n \to \mathbb{R}^q$ represents inequality constraints. The solution $x^\star$ is optimal if it minimizes the objective while satisfying all constraints.

For finite-horizon setpoint optimization, the decision vector contains the control actions for several time steps. The objective is usually the accumulated operating cost over the horizon. The constraints describe system dynamics, energy balances, storage bounds, and terminal requirements. This time-coupled structure is important because a state such as battery state of charge or thermal storage temperature depends on previous decisions.

The difficulty of an optimization problem depends strongly on its mathematical structure. Linear and mixed-integer linear formulations are widely used in energy-system applications because simplified models can often be solved reliably. However, realistic energy conversion systems frequently include nonlinear component behavior, continuous control variables, and coupled dynamic constraints. Such systems lead to nonlinear programming (NLP) formulations, which are generally harder to solve and may contain local optima [4]. As the number of time steps, decision variables, nonlinear component models, and coupling constraints increases, the optimization problem generally becomes larger and more difficult to solve within a short operational time window [2].

Another important distinction is whether the problem is deterministic or stochastic. In deterministic optimization, all relevant parameters, such as electricity prices, wind availability, and demand, are assumed to be known over the optimization horizon. Stochastic optimization represents uncertain quantities through random variables or scenarios. This is closer to real operation, but it also increases model size and computational complexity.

Nonlinear constrained problems are usually solved with dedicated numerical solvers. Interior-point methods are common in this setting. IPOPT, for example, is an open-source solver for large-scale nonlinear programming and uses a primal-dual interior-point method with a filter line-search strategy [5]. These solvers can be very effective when the model is well defined, but their performance depends on model accuracy, constraint structure, scaling, and initialization. Such sensitivity is one reason hybrid strategies, such as the Bayesian-optimization and IPOPT approach of Kyriakidis et al., have been proposed for complex nonconvex energy-conversion problems [2].

2.3 Reinforcement Learning

Introduces reinforcement learning as a data-driven approach for learning optimal control policies through interaction with the environment.

Reinforcement learning (RL) offers a different view of the same setpoint problem. Instead of solving an optimization problem from scratch at every decision time, an

agent learns from repeated interaction with an environment. At each time step, the agent observes the current state, selects an action, receives a reward, and observes the next state. Over time, it adapts its policy so that actions leading to high long-term reward become more likely [6].

Formally, RL problems are commonly modeled as a Markov Decision Process (MDP), defined by the tuple

$$(\mathcal{S}, \mathcal{A}, p, r, \gamma_{\mathrm{RL}}),$$

where

- $\mathcal{S}$ is the set of states representing the system,
- $\mathcal{A}$ is the set of actions,
- $p(s' \mid s, a)$ defines the transition dynamics,
- $r(s, a)$ is the reward function,
- $\gamma_{\mathrm{RL}} \in [0, 1]$ is the discount factor.

The Markov property means that the current state contains the information needed to describe the probability distribution of the next state. A policy $\pi(a \mid s)$ describes how the agent selects actions. The return from time step k is the discounted cumulative reward

$$G_k = \sum_{i=0}^{N-k-1} \gamma_{\mathrm{RL}}^i r(s_{k+i}, a_{k+i}),$$

and the learning objective is to find a policy that maximizes the expected return. Value functions estimate the expected return from a state, while action-value functions estimate the expected return from a state-action pair. These estimates are important because they allow the agent to consider the future consequences of a decision, not only its immediate reward [6].

In energy-system applications, the decision problem can often be interpreted as stochastic optimal control. Uncertainty enters through external variables such as renewable generation, electricity prices, and demand. These variables can be modeled as stochastic processes, which leads to a finite-horizon MDP formulation [3]. Since the engineering objective is usually cost minimization, the reward is often defined as the negative operating cost. The resulting goal is to minimize expected operating cost over time while respecting system dynamics and constraints.

Classical dynamic programming can solve some MDPs through backward recursion. However, this approach becomes impractical when the state or action space grows, a difficulty known as the curse of dimensionality. Realistic energy systems often have continuous states, continuous actions, and long horizons, so exact dynamic-programming solutions are usually not tractable.

RL addresses this limitation by learning approximate policies from interaction data. A trained policy can then compute an action directly from the current observation. In model-free RL, the learning algorithm does not require an explicit analytical transition model for planning, although training still requires interaction data. In engineering applications, this data is usually generated with a simulator or digital model, so the quality of the learned policy depends strongly on the quality of the training environment.

To locate the algorithm used in this thesis, it is helpful to briefly classify the main families of RL algorithms. RL methods are not separated by one single list of

types; instead, they are usually described through several overlapping distinctions [6]:

- Model-based and model-free methods: model-based methods learn or use a model of the environment dynamics for planning. Model-free methods learn directly from sampled transitions.
- Value-based and policy-based methods: value-based methods learn value or action-value functions and choose actions from them. Policy-based methods directly optimize a parameterized policy.
- Actor-critic methods: actor-critic methods combine both ideas. The actor represents the policy, and the critic estimates values or action-values used to improve that policy.
- On-policy and off-policy methods: on-policy methods learn from data generated by the current policy. Off-policy methods can also reuse data generated by previous policies or exploratory behavior.
- Tabular and deep RL methods: tabular methods store values explicitly for small state and action spaces. Deep RL uses neural networks as function approximators and is therefore more suitable for large or continuous spaces.

The choice of algorithm follows from the structure of the problem. In this thesis, the relevant setpoints are continuous quantities, such as battery power, temperature levels, mass flow rates, and bypass positions. Discretizing these variables would either remove useful operating points or create a very large action space. A suitable algorithm should therefore learn a parameterized policy and output continuous actions directly.

Policy-gradient and actor-critic methods are commonly used for such problems because they optimize a parameterized policy instead of enumerating all possible actions [3]. In an actor-critic setting, the actor can propose continuous setpoints, while the critic evaluates their expected long-term effect on cost and future system states. Within the classification above, Soft Actor-Critic (SAC) is a model-free, off-policy, actor-critic, deep reinforcement-learning algorithm.

SAC is used in this thesis because it extends the model-free actor-critic framework with a maximum-entropy objective for continuous-control problems [7]. This is useful for the present application because the agent must search through continuous operating setpoints during training, while still learning a policy that can be evaluated quickly after training.

In standard RL, the policy is usually trained to maximize expected return. SAC adds an entropy term to this objective. Intuitively, this means that the policy is encouraged to keep exploring when several actions appear useful, instead of becoming too deterministic too early. The objective can be interpreted as maximizing both reward and stochasticity:

$$\mathbb{E}_{\pi}\left[\sum_{k=0}^{N-1}\gamma_{\mathrm{RL}}^{k}\left(r(s_k,a_k)+\alpha\mathcal{H}(\pi(\cdot\mid s_k))\right)\right],$$

where α controls the weight of the entropy term and $\mathcal{H}$ denotes policy entropy [7]. This entropy term can improve exploration and reduce the risk that the agent converges too early to a narrow operating strategy.

SAC is therefore a suitable choice for the systems considered in this thesis. It can handle continuous action spaces, reuse past experience through off-policy learning, and support exploration in environments with time-varying prices, renewable generation, and storage dynamics.

3 Use Cases

Describes the two systems considered in this thesis and their operational challenges.

3.1 Battery Energy Storage System

Describes the battery system used as the first case study.

Battery energy storage systems (BESS) are electrochemical devices that store electrical energy for later use and provide grid services such as energy arbitrage, peak shaving, and short-term balancing. A typical BESS is characterized by its usable energy capacity, maximum charge/discharge power, round-trip efficiency, and state-of-charge (SOC) operating limits. Real-world operation must balance economic objectives (e.g., buying low / using stored energy when prices are high) with engineering constraints (SOC bounds, power limits, efficiency losses).

BESS presents several challenges. Decisions are inherently time-coupled because current charging/discharging changes future availability. In addition, uncertain exogenous signals (market prices) require policies that are robust to forecast errors, and hard feasibility constraints must be enforced at every time step. These properties make BESS an illustrative case study for comparing optimization-based and reinforcement-learning approaches to real-time setpoint determination in energy systems.

A battery energy storage system provides flexible, time-shiftable power by charging when electricity prices are low and discharging when prices are high to reduce grid procurement cost. In this thesis, the battery is modeled using the `BatteryEnv` environment: the agent's observation includes the state-of-charge (SOC), a normalized current price, a normalized time index, and a forecast window of upcoming prices; the action is a continuous charge/discharge command $a \in [-1, 1]$ scaled to real power $P \in [-P_{\max}, P_{\max}]$.

The SOC dynamics are modeled as

$$\mathrm{SOC}_{t+1} = \mathrm{SOC}_t + \frac{P_{\mathrm{actual},t}\,\Delta t}{E_{\max}},$$

where $P_{\mathrm{actual},t}$ is the effective storage-side power after efficiency losses, $E_{\max}$ the usable energy capacity, and Δt the time step. The implemented environment uses the sign convention $P_t > 0$ for charging and $P_t < 0$ for discharging, with a simplified symmetric efficiency coefficient η. The system is subject to the following operational constraints:

$$0 \le \mathrm{SOC}_t \le 1, \quad \forall t, \tag{2}$$

$$-P_{\max} \le P_t \le P_{\max}, \quad \forall t, \tag{3}$$

$$P_{\mathrm{actual},t} = \eta P_t, \quad \forall t. \tag{4}$$

where $P_{\max}$ is the maximum charge/discharge power. Constraint (2) ensures the battery operates within safe operating levels, Constraint (3) enforces physical power

limitations, and Constraint (4) models the efficiency losses used in the RL environment. The environment enforces feasibility by dynamically limiting the commanded power at each step so that the SOC remains within bounds.

The goal is to reduce the cost of electricity from the grid, which is calculated as the price multiplied by the power used. The environment gives the negative of this cost as the reward. In this work, there is no on-site generation and no selling back to the grid. The battery charges from the grid when prices are low and discharges to serve local demand when prices are high. Main challenges include respecting SOC and power limits, accounting for efficiency losses, handling uncertain future prices, and managing the fact that actions now affect future energy availability. These features make the BESS a suitable test case for reinforcement learning, where a policy must balance using energy now against saving it for likely higher-price periods while always obeying feasibility limits.

3.2 Cookie Production Factory

Introduces the industrial production system considered as the second case study.

The second use case represents a non-linear cookie production factory with an electrified steam-generation utility system. The production process requires a constant supply of process heat in the form of superheated steam, while the utility system must decide how to operate the energy conversion components under fluctuating wind generation and electricity prices. The operational objective is to satisfy the steam demand while minimizing grid electricity cost over a finite scheduling horizon. The system model is based on the industrial power-to-heat system introduced by Walden et al. [1] and later used as a renewable steam-generation benchmark by Kyriakidis et al. [2].

The system consists of four main units: a wind turbine (WT), a closed reverse Brayton-cycle high-temperature heat pump (HTHP), a sensible thermal energy storage (TES), and a steam generator (SG). The WT supplies renewable electricity, while additional electricity can be purchased from the grid. The HTHP uses a waste-heat air stream as heat source and delivers high-temperature process heat to an intermediate thermal-oil loop. The thermal oil acts as heat transfer fluid and can be routed through the TES or directly to the SG by controllable bypasses. The SG provides the required steam conditions for the consumer factory. The HTHP and SG are represented by polynomial surrogate models, while the TES is modeled as a simplified single energy reservoir, and the charging/discharging behavior is represented using efficiency equations.

The corresponding finite-horizon optimization problem is formulated over a dis-

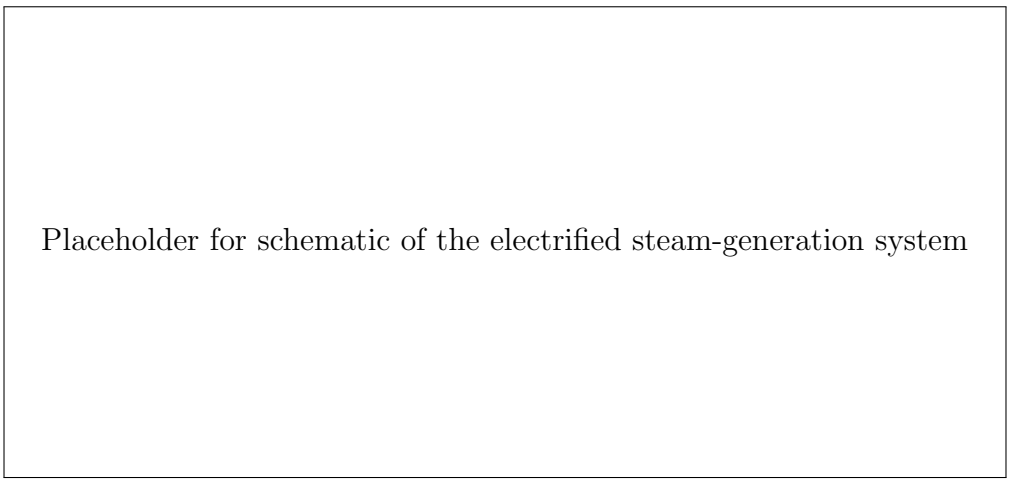


Figure 1: Schematic representation of the electrified steam-generation system serving the cookie production factory, adapted from the industrial power-to-heat system introduced by Walden et al. [1] and the renewable steam-generation benchmark described by Kyriakidis et al. [2].

crete time horizon with time index $k \in \{1, \ldots, n\}$:

$$\min_{\mathbf{x}} \quad J = \sum_{k=1}^{n} P_{\text{grid}}^{k} g_{\text{grid}}^{k} \Delta t, \tag{5}$$

$$\text{s.t.} \quad P_{\text{grid}}^{k} + P_{\text{WT}}^{k} = 3F_{\text{HTHP}}\left(T_{\text{I}}^{k}, \dot{m}_{\text{I}}^{k}, T_{\text{III}}^{k}, R^{k}\right), \tag{6}$$

$$T_{\text{II}}^{k} = F_{\text{HTHX}}\left(T_{\text{I}}^{k}, \dot{m}_{\text{I}}^{k}, T_{\text{III}}^{k}, R^{k}\right), \tag{7}$$

$$T_{\text{IV}}^{k} = F_{\text{LTHX}}\left(T_{\text{I}}^{k}, \dot{m}_{\text{I}}^{k}, T_{\text{III}}^{k}, R^{k}\right), \tag{8}$$

$$T_{2}^{k} = \beta_{1}^{k} T_{\text{II}}^{k} + \left(1 - \beta_{1}^{k}\right) T_{1}^{k}, \tag{9}$$

$$T_{\text{I}}^{k} = \beta_{2}^{k} T_{3}^{k} + \left(1 - \beta_{2}^{k}\right) T_{4}^{k}, \tag{10}$$

$$T_{2}^{k} = 201.92 + \frac{1819.32}{3\dot{m}_{\text{II}}^{k}}, \tag{11}$$

$$T_{3}^{k} = 196.3 - \frac{188.4}{3\dot{m}_{\text{I}}^{k}}, \tag{12}$$

$$\dot{Q}_{s,\text{ch}}^{k} = 3\dot{m}_{\text{II}}^{k} c_{p,\text{f}}\left(T_{\text{II}}^{k} - T_{1}^{k}\right)\left(1 - \beta_{1}^{k}\right), \tag{13}$$

$$\dot{Q}_{s,\text{dch}}^{k} = 3\dot{m}_{\text{I}}^{k} c_{p,\text{f}}\left(T_{4}^{k} - T_{3}^{k}\right)\left(1 - \beta_{2}^{k}\right), \tag{14}$$

$$\dot{Q}_{s,\text{ch}}^{k} \dot{Q}_{s,\text{dch}}^{k} \leq \gamma, \tag{15}$$

$$T_{1}^{k} = T_{\text{II}}^{k} - \epsilon_{\text{ch}}\left(T_{\text{II}}^{k} - T_{s}^{k-1}\right), \tag{16}$$

$$T_{4}^{k} = T_{3}^{k} - \epsilon_{\text{dch}}\left(T_{3}^{k} - T_{s}^{k-1}\right), \tag{17}$$

$$T_{s}^{k} = T_{s}^{k-1} + \frac{\dot{Q}_{s,\text{ch}}^{k} - \dot{Q}_{s,\text{dch}}^{k}}{m_{s} c_{p,\text{s}}} \Delta t, \tag{18}$$

$$T_{s}^{0} = \tilde{T}_{0}, \qquad T_{s}^{n} = T_{s}^{0}, \tag{19}$$

$$T_{\text{I}}^{k} \in [177, 250], \qquad \dot{m}_{\text{I}}^{k} \in [5, 16], \tag{20}$$

$$T_{\text{III}}^{k} \in [60, 100], \qquad R^{k} \in [0.8, 1.53], \qquad \beta_{1}^{k}, \beta_{2}^{k} \in [0, 1]. \tag{21}$$

Here, the objective in Eq. (5) minimizes the electricity cost of grid power consumption. The power balance in Eq. (6) couples grid electricity, wind power, and HTHP demand. The remaining constraints describe the HTHP and SG surrogate models, bypass mixing, TES charging and discharging, the prohibition of simultaneous

charge and discharge, cyclic storage operation, and the admissible operating ranges of the main decision variables.

The main control decisions are the bypass variables β_1 and β_2, the compressor-speed-related variable R, the mass flow rate $\dot{m}_{\mathrm{I}}$, and the HTHP inlet temperature T_{I}. The bypass variables determine the operating mode of the system. In charging mode, heat from the HTHP is routed into the TES; in discharging mode, stored heat is released to support the steam-generation process; and in idle or bypass operation, the storage is not actively charged or discharged. Simultaneous charging and discharging is prohibited through a complementarity constraint.

In the reinforcement-learning environment, the agent observes the normalized TES temperature, current electricity price, available wind power, the previous operating mode, and forecast information for price and wind availability. The action is a continuous vector containing β_1, β_2, R, $\dot{m}_{\mathrm{I}}$, and T_{I}. The reward is based on the negative grid-electricity cost, with additional penalties for violating operational requirements such as TES temperature bounds, invalid mode behavior, coupling residuals, and deviation from the cyclic terminal storage condition.

This use case is more complex than the battery system because it combines nonlinear surrogate equations, thermal storage dynamics, multiple coupled continuous control variables, and hard process constraints. It therefore provides a suitable second benchmark for evaluating whether reinforcement learning can learn feasible and cost-effective setpoint policies for industrial energy utility systems, and for comparing the learned policy against the existing mathematical optimization formulation.

4 Methodology

Describes how the optimization problem is formulated as a reinforcement learning problem.

The previous section introduced the two energy utility systems considered in this thesis and described their operational objectives and constraints. This section explains how these system-specific optimization problems are reformulated as reinforcement-learning problems. In particular, the setpoint optimization task is expressed as a finite-horizon sequential decision problem, where an agent observes the current system state, selects continuous control actions, and receives a reward based on operating cost and constraint satisfaction.

4.1 Problem Formulation

Formulates the system control problem as a reinforcement learning problem.

For both use cases, the objective is to determine operating setpoints over a discrete time horizon such that the total operating cost is minimized while all relevant constraints are respected. Let $k \in \{0, \dots, N-1\}$ denote the discrete time index. At each time step, the system is described by a state vector s_k, the controller selects an action vector a_k, and the system evolves to the next state s_{k+1} according to the underlying system dynamics and exogenous inputs such as electricity prices and renewable generation.

The resulting decision problem is formulated as a Markov decision process (MDP)

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, p, r, \gamma_{\mathrm{RL}}),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $p(s_{k+1} \mid s_k, a_k)$ denotes the transition dynamics, $r(s_k, a_k)$ is the reward function, and γ_{RL} is the discount factor.

The agent aims to learn a policy $\pi(a_k \mid s_k)$ that maximizes the expected cumulative reward [6]:

$$\max_{\pi} \ \mathbb{E}_{\pi}\left[\sum_{k=0}^{N-1} \gamma_{\mathrm{RL}}^{k} r(s_k, a_k)\right].$$

Since the original engineering objective is cost minimization, the reward is defined as the negative operating cost, augmented by penalty terms for constraint violations where required. In this way, feasible and economically favorable operation corresponds to a high cumulative reward. The following subsections describe how this general formulation is implemented in the reinforcement-learning framework, training setup, and evaluation procedure.

4.2 Reinforcement Learning Framework

Explains the structure of the RL agent and learning algorithm.

The reinforcement-learning framework used in this thesis follows the same structure for both use cases. Each physical system is represented by a simulation environment with a `reset` function for initializing an episode and a `step` function for applying control action(s). At time step k, the environment returns an observation s_k, the SAC agent samples or predicts an action a_k, and the environment maps this action to admissible physical setpoints before computing the next state, reward, and termination signal. A transition is therefore generated as

$$\tau_k = (s_k, a_k, r_k, s_{k+1}, d_k),$$

where d_k indicates whether the episode has ended. These transitions are stored in a replay buffer and reused during training, which is the off-policy part of SAC.

The policy is represented by a neural network $\pi_\theta(a_k \mid s_k)$ that outputs continuous setpoints. This is important for the considered systems because the relevant decisions are not naturally discrete: battery power, bypass positions, mass flow rates, compressor settings, and temperatures can vary continuously within their operating ranges. The critic is represented by action-value networks $Q_{\phi_i}(s_k, a_k)$, which estimate the expected future return of a state-action pair. SAC uses two critics and target networks to reduce overestimation bias and to stabilize the bootstrapped value estimates [7].

During training, an action sampled from the stochastic actor is first processed by an environment-specific action map

$$a_k^{\mathrm{phys}} = \mathcal{R}(s_k, a_k),$$

where $\mathcal{R}$ converts the normalized policy output into physical setpoints and applies feasibility handling. For the battery system, this map scales the scalar action to charge or discharge power and limits the effective power near the SOC bounds. Thus, the RL policy proposes setpoints, while the environment remains responsible for enforcing engineering constraints.

The reward has the same general form in both environments:

$$r_k = -C_k - \sum_{\ell} \lambda_\ell v_{\ell,k} - \lambda_{\mathrm{T}} v_{\mathrm{T},k},$$

where C_k is the grid-electricity cost at time step k, $v_{\ell,k}$ are constraint-violation terms, and λ_ℓ are penalty weights. The terminal term $v_{\mathrm{T},k}$ is only active at the end

of an episode. In the battery case it penalizes an insufficient terminal SOC, while in the cookie factory it penalizes deviation from the cyclic terminal TES temperature. This reward design converts the original cost-minimization problem into a return-maximization problem without removing the need for physical feasibility.

The SAC update is based on mini-batches sampled from the replay buffer $\mathcal{D}$. For a sampled transition, the critic target is computed using the reward, the discounted target critic value, and the entropy term:

$$y_k = r_k + \gamma_{\mathrm{RL}}(1 - d_k)\left[\min_i Q_{\bar{\phi}_i}(s_{k+1}, a_{k+1}) - \alpha \log \pi_\theta(a_{k+1} \mid s_{k+1})\right],$$

with $a_{k+1} \sim \pi_\theta(\cdot \mid s_{k+1})$. The critic parameters are updated by minimizing the squared difference between $Q_{\phi_i}(s_k, a_k)$ and y_k. The actor is updated to choose actions with high expected value while retaining entropy:

$$J_\pi(\theta) = \mathbb{E}_{s_k \sim \mathcal{D},\, a_k \sim \pi_\theta}\left[\alpha \log \pi_\theta(a_k \mid s_k) - \min_i Q_{\phi_i}(s_k, a_k)\right].$$

The temperature parameter α controls the trade-off between exploitation and exploration. A larger entropy contribution encourages broader exploration of the continuous setpoint space, while the critic gradually guides the actor toward actions that reduce cost and avoid constraint penalties.

After training, the stochastic exploration behavior is no longer required for comparing schedules. The learned policy is therefore evaluated deterministically by selecting the mean action predicted by the actor at each time step. This produces one reproducible setpoint trajectory for a given price and wind profile and allows direct comparison with the mathematical optimization baseline in terms of cost, constraint satisfaction, and computation time.

4.3 Training Setup

Describes the training environment, reward function, and hyperparameter configuration.

Training is performed in reinforcement-learning environments before the learned policies are evaluated on fixed test trajectories. This is necessary because direct trial-and-error interaction with real energy utility systems would be unsafe and costly. The environment provides the transition dynamics, cost calculation, constraint handling, and terminal condition for each episode. The learning algorithm only receives observations, rewards, and termination signals through the environment interface.

For both use cases, one episode corresponds to a 24-step scheduling horizon. During training, the start index of the episode is sampled randomly from the available time-series data. This random-window strategy exposes the agent to different price and renewable-generation patterns instead of repeatedly starting each episode at the same time of day. Both environments use the same grid-price series; the cookie factory environment additionally samples the corresponding wind-availability values for the same 24-hour window. The forecast horizon is also set to 24 steps, so the observation contains the current operating state together with information about upcoming exogenous signals.

The training environments are wrapped with monitoring and vectorized environment interfaces. Monitoring records episode rewards and lengths for later inspection, while the vectorized wrapper provides the interface expected by the Stable-Baselines3 implementation of SAC. Only one environment instance is used at a time

in the current setup, so the vectorization is mainly an interface layer rather than a parallel simulation strategy.

The BESS reward is the negative grid-side electricity cost at each time step, with an additional terminal penalty if the final SOC falls below the desired terminal level. The cookie factory reward also starts from the negative grid-electricity cost, but includes additional penalties for different constraint violations. The penalty terms are therefore more important in the second use case, because feasibility cannot be represented by simple action bounds alone.

The main SAC hyperparameters used in the implementation are summarized in Table 1. The same learning-rate schedule and discount factor are used for both use cases to provide a consistent baseline configuration for continuous-control learning. Differences between the two training setups mainly arise from the different state variables, action spaces, and constraint structures of the two environments.

Parameter	Battery system	Cookie factory
Algorithm	SAC	SAC
Policy	MLP policy	MLP policy
Replay buffer size	200000	200000
Batch size	256	256
Learning rate	linear schedule from $3 \cdot 10^{-4}$	linear schedule from $3 \cdot 10^{-4}$
Discount factor γ_{RL}	0.99	0.99
Target smoothing coefficient τ	0.005	0.005
Training frequency	1 step	1 step
Gradient steps per update	1	1
Episode length	24 steps	24 steps
Forecast horizon	24 steps	24 steps

Table 1: Main SAC training settings used for the two reinforcement-learning environments.

For the both battery system and cookie factory, the policy and critic networks use fully connected neural networks with two hidden layers of 256 neurons and ReLU activation functions. In both cases, trained models are saved after learning and are later loaded for deterministic evaluation. This separation between training and evaluation is important: stochastic actions are useful during learning, but deterministic actions are used for the final comparison with mathematical optimization.

4.4 Evaluation Metrics

The trained RL policies are evaluated through deterministic rollouts on fixed 24-hour evaluation windows. Deterministic evaluation means that the actor returns one action for each observation without sampling exploration noise. This makes the resulting setpoint trajectory reproducible and allows a direct comparison with the mathematical optimization baseline on the same grid-price and wind-availability profiles.

The primary economic metric is the total grid-electricity cost over the evaluation horizon:

$$J_{\mathrm{grid}} = \sum_{k=0}^{N-1} C_k,$$

where C_k is the grid-electricity cost at time step k. For the battery system, this cost is computed from the electricity price and the battery power exchanged with the grid. For the cookie factory, it is computed from the grid power required by the heat-pump system after accounting for available wind power. Since the engineering objective is cost minimization, a lower value of J_{grid} indicates better economic performance.

The cumulative reward is reported as a secondary metric:

$$R_{\text{ep}} = \sum_{k=0}^{N-1} r_k.$$

This value is useful for analyzing the RL objective because it combines operating cost with penalty terms. However, it is not used as the only comparison criterion, since the reward may include artificial penalty weights that are not part of the physical operating cost. Therefore, total cost and constraint satisfaction are evaluated separately.

Constraint satisfaction is measured by aggregating the violation terms recorded during the rollout. For a set of constraint residuals or violation magnitudes $v_{\ell,k}$, the total violation is defined as

$$V_{\text{tot}} = \sum_{k=0}^{N-1} \sum_{\ell} v_{\ell,k}.$$

In addition, the number or frequency of time steps with nonzero violations is considered:

$$f_{\text{viol}} = \frac{1}{N} \sum_{k=0}^{N-1} \mathbb{I} \left(\sum_{\ell} v_{\ell,k} > \varepsilon \right),$$

where ε is a small numerical tolerance. For the battery system, the relevant checks include SOC bounds, power limits, and the terminal SOC requirement. For the cookie factory, the violation categories include bypass feasibility, complementarity of charging and discharging, TES temperature bounds, grid-power bounds, coupling residuals, and the terminal cyclic TES condition.

The evaluation also records trajectory-level operating indicators. For the BESS, these include the SOC trajectory, commanded and effective battery power, terminal SOC, and cumulative cost.

Finally, computation time is used to assess practical applicability. For mathematical optimization, the relevant value is the solver time required to compute a complete schedule for the evaluation horizon. For RL, the one-time training effort is reported separately from the online evaluation time. After training, the relevant online cost is the time required for a policy forward pass and one environment step at each decision time. This distinction is important because RL can require substantial training effort, but a trained policy can usually produce setpoints much faster than solving a nonlinear optimization problem from scratch for every horizon.

5 Results and Discussion

Presents and analyzes the experimental results.

5.1 Battery Energy Storage System

5.2 Cookie Production Factory

5.3 Key Findings

6 Conclusion and Future Work

Summarizes the main findings and outlines possible future work.

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